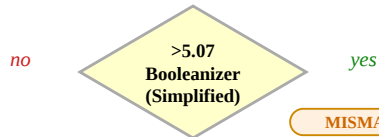


Booleanization

Raw Features: int/float

F_1 1.33 F_2 7.25



Boolean Features: 0/1

B_1 0 B_2 1

Boolean Feature + Complement

Boolean Literals: 0/1

B_1 0 $\bar{B}_1$ 1 B_2 1 $\bar{B}_2$ 0

INPUTS

MISMATCH DETECTION

RESULT

$c = \text{count_ones}(\text{val}) = 1$ mismatch
 $\text{clamp} = \min(\text{lit_sum}=3, \text{LF}=4) = 3$
 $\text{output} = \max(\text{clamp} - c, 0) = \max(3-1, 0) = 2 \in [0, \text{LF}]$

Fuzzy Clause Computation

$\text{check_clause}(x, \text{literals}, \text{literals_inv}, \text{clamp})$

	B_1	$\bar{B}_1$	B_2	$\bar{B}_2$	...
$x.\text{chunks}[ij]$	0	1	1	0	
$\text{literals}[ij]$	1	1	1	0	
$\text{literals_inv}[ij]$	0	0	0	1	

1 = included
0 = excluded

$(\sim x \ \& \ \text{lit})$	1	0	0	0	
$(x \ \& \ \text{lit})$	0	0	0	0	

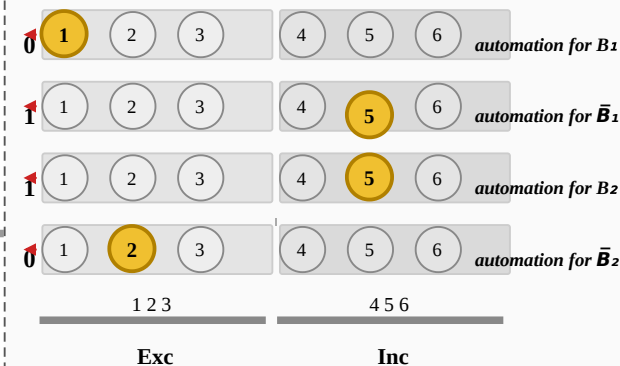
lit=1, x=0 (absent)
lit=1, x=1 (unexpected)

$\text{val} = \text{op}_1 \mid \text{op}_2$

1 0 0 0

mismatch
match

Tsetlin Automata States after training



$\text{state} \geq \text{include_limit} \rightarrow \text{bit}=1 \text{ in } \text{literals}[]$
 $\text{state} < \text{include_limit} \rightarrow \text{bit}=0$

Multiclass Fuzzy Pattern Tsetlin Machine Inference

